

Sospeter Mburu Kinyanjui

Software Engineer

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Summary

Backend engineer specializing in Go and Rust, focused on building systems people can rely on: payments that settle, apps that keep working without internet, and services that stay secure under load. I take ownership from architecture to deployment, and have shipped hackathon and personal products end-to-end across fintech, logistics, mobility and developer tooling for East African users. Comfortable defending technical decisions in peer review, writing supporting documentation, and working with Docker, CI/CD and infrastructure as code.

Languages: Go, Rust, TypeScript, Python, C, C#

Top projects

Kyla-2FA

- Built an offline desktop two-factor authenticator with local-only storage, giving users an alternative to cloud-based authenticator apps.
- Implemented RFC 6238-compliant TOTP generation and encrypted local storage to manage authenticator entries with no network dependency.

Skill-match

- Built an AI-powered job-search assistant with persistent agentic memory for the CockroachDB × AWS hackathon, so the assistant retains context about each job seeker across sessions.
- Migrated the data layer from CockroachDB to PostgreSQL: pgvector serves as the memory layer, Google Gemini powers the AI, and resume files are stored on the server's local filesystem.

emrent

- Implemented the BitTorrent (BEP 3) and DHT (BEP 5) specifications from scratch as a Cargo workspace of 5 independently unit-tested layers (bencode, DHT peer discovery, metainfo parsing, peer wire protocol, piece verification).
- Implemented async peer wire handling on Tokio supporting 150+ concurrent peer connections per torrent, with a Tauri desktop UI for torrent management.

Experience

Zone01 Kisumu — Software Engineering

2026 — Present

- Project-based, peer-to-peer model: skills are built by shipping real projects, defending technical decisions in peer code review, and giving structured feedback on others' work. All 6 projects below were built here, 3 of them as hackathon entries.
- **EcoRoute** (hackathon): smart waste collection for cleaner cities. Monitors collection points, optimizes truck routes, lets residents report issues, and measures the environmental impact of every run.
- **SkyFee** (hackathon): Kenyan school-fee payments over Bitcoin Lightning. Parents complete a 4-step flow (pick partner school, verify admission number, pay a KES-denominated invoice, track settlement) ending in BTC-to-KES off-ramp and M-Pesa payout.
- **Duka Imara** (hackathon): PWA for market vendors and spaza/tuck shops with unreliable connectivity and shared devices. Runs 100% offline with 0 data lost, and syncs automatically when connectivity returns.
- **Ride-hailing platform** (East Africa, Uganda & Kenya): connects travelers with verified taxi-association drivers across 2 countries. Every driver belongs to a known SACCO and is admin-verified; every rider is verified through 1 of 2 mobile operators (MTN Uganda / Safaricom Kenya) before booking.
- **forum**: Go web forum with posts, comments, categories, and likes/dislikes; secured its 14-endpoint REST API with custom auth and CORS middleware, with 0 unauthorized-access issues found across 50+ test accounts in review.
- **traceroute-te** (personal): axum-backed web reimplement of the Linux traceroute utility.